

Exact strength of the subset-divisibility obstruction

Research notes, 30 September 2026

Status. The finite statements below have complete elementary proofs. They concern necessary arithmetic conditions for permutation-fair dice, not the existence of dice with prescribed face counts. No claimed resolution of either principal size question follows.

Write $\vartheta(x) = \sum_{p \leq x} \log p$, with sums over primes, and put

$$P_n = \prod_{p \leq n} p^{n-p+1} = \exp \left(\sum_{j=1}^n \vartheta(j) \right).$$

Call a positive integer vector $c = (c_1, \dots, c_n)$ arithmetically admissible if

$$|S|! \mid \prod_{i \in S} c_i \quad \text{for every } S \subseteq [n].$$

Every permutation-fair face-count vector is arithmetically admissible.

Theorem 1 (Complete characterization). *The following conditions on c are equivalent:*

1. c is arithmetically admissible;
2. for every prime $p \leq n$, at most $p - 1$ coordinates of c are not divisible by p .

Thus restrictions to prime-sized subsets already imply all the subset-factorial divisibility conditions. Higher prime powers supply no additional obstruction of this form.

Proof. Necessity follows by taking a set of p coordinates not divisible by p . For sufficiency, fix S of size k and a prime $p \leq k$. At least $k - p + 1$ coordinates in S are divisible by p . Legendre's formula gives

$$v_p(k!) = \frac{k - s_p(k)}{p - 1} \leq \frac{k - 1}{p - 1} \leq k - p + 1,$$

where $s_p(k) \geq 1$ is the base- p digit sum; the last inequality is equivalent to $(p - 2)(k - p) \geq 0$. Hence $v_p(\prod_{i \in S} c_i) \geq v_p(k!)$. This holds for every relevant prime. $\square$

Theorem 2 (Sharp product obstruction, nearly sharp length obstruction). *Among arithmetically admissible vectors, the exact minimum of $\prod_{i=1}^n c_i$ is P_n . If*

$$B_n = \min \left\{ \sum_i c_i : c \text{ arithmetically admissible} \right\},$$

then

$$nP_n^{1/n} \leq B_n \leq n^2 P_n^{1/n}.$$

Moreover, an admissible squarefree vector attaining product P_n can be constructed greedily with $\max_i c_i / \min_i c_i \leq n$.

Proof. The characterization forces each prime $p \leq n$ into at least $n - p + 1$ coordinates, so every product is divisible by P_n . Start all coordinates at 1. For each prime $p \leq n$, multiply the $n - p + 1$ currently smallest coordinates by p . Each prime occurs in exactly the required number of coordinates and nowhere with exponent greater than 1, so the final product is P_n and the vector is admissible.

If the old coordinates have maximum/minimum ratio at most R , the new ratio is at most $\max(R, p)$. Indeed, for two coordinates that are both multiplied or both left alone, their ratio is unchanged. The ratio of a multiplied coordinate to an unmultiplied coordinate is at most p , since all multiplied coordinates were among the smallest. The reverse ratio is at most $R/p \leq R$. Induction proves the asserted ratio bound. Therefore every coordinate is at most $nP_n^{1/n}$, giving the upper bound for B_n ; the lower bound is AM–GM. $\square$

Corollary 3 (Asymptotic barrier). *The prime number theorem gives*

$$\log P_n = \left(\frac{1}{2} + o(1)\right) n^2, \quad B_n = \exp\left(\left(\frac{1}{2} + o(1)\right) n\right).$$

Consequently every permutation-fair collection has

$$\prod_i c_i \geq P_n, \quad \max_i c_i \geq P_n^{1/n}, \quad \sum_i c_i \geq nP_n^{1/n}.$$

No argument using only subset-factorial divisibility can improve the coefficient 1/2 in the exponential growth of the maximum or total number of faces.

Proof. Use $\log P_n = \sum_{j=1}^n \vartheta(j)$ and $\vartheta(j) = j + o(j)$. The explicit balanced admissible vectors prove the barrier statement. $\square$

For equal coordinates, arithmetic admissibility is equivalent to $\prod_{p \leq n} p \mid c_1$. Hence the exact equal-coordinate arithmetic optimum is $n \exp(\vartheta(n)) = \exp((1 + o(1))n)$. The arithmetic relaxation alone therefore exhibits a ratio $\exp((1/2 + o(1))n)$ between its equal and arbitrary optima. This is compatible with a polynomial relation between actual dice optima, but cannot establish one.

Corollary 4 (Restriction quantiles). *If the face counts of a permutation-fair collection are arranged as $c_{(1)} \leq \dots \leq c_{(n)}$, then, for every $1 \leq r \leq n$,*

$$\prod_{i=1}^r c_{(i)} \geq P_r, \quad c_{(r)} \geq P_r^{1/r}.$$

In particular, as $r \rightarrow \infty$, $c_{(r)} \geq \exp((1/2 + o(1))r)$.

Remark 5 (Relation to existing literature). *Purcell’s Proposition 1 records the equal-sided primordial obstruction: <https://doi.org/10.1080/00029890.2024.2345576>. The manuscript already proves that at least $n - p + 1$ face counts must be divisible by p . The new content here is the converse for the entire arithmetic relaxation, its exact product optimum, and the matching balanced-vector construction. Novelty beyond these checked sources has not been established by an exhaustive literature search.*